Joseph Ukeje

Senior Software Engineer | Distributed Systems, Payments & AI

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Summary

Senior Software Engineer with **7 years of experience** building transaction-heavy products across payments, core banking, enterprise platforms and AI automation. Designs **distributed, event-driven architectures across AWS and Azure** and owns services from technical design through implementation, testing, deployment and production recovery. Experience spans platforms processing **5–10 million transactions daily**, reliable messaging, financial correctness and reviewable **generative-AI integrations**. Collaborates with Product, QA, DevOps/SRE, Security and Operations, with code review, mentoring and evidence-based incident investigation.

Core Competencies

Distributed systems | Event-driven architecture | API design | Idempotency & concurrency control | Payments, settlement & reconciliation | Production ownership | Security & compliance | AI integration | Technical leadership

Professional Experience

Senior Software Engineer *Apr 2025 - Present*

Interswitch Group Remote

Core Banking & Payments Platform

- Architected a **centralised audit service** with **Kafka-backed asynchronous ingestion**, capturing immutable trails for **5–10 million inbound and outbound transactions daily** and user actions so Compliance and Operations could trace financial activity across distributed services.
- Built **IRIS, a core-banking payment gateway and inter-bank transfer engine**, with TCP connection management and **gRPC/Protobuf** contracts. Delivered **ISO 20022/SWIFT pacs.008, pacs.002 and pacs.028** flows with canonicalisation, signatures and recovery, reducing failures by **98% in the affected payment flows**.
- Implemented **ISO 8583 card-payment message processing** as a separate flow, validating message fields and mapping requests and responses into auditable transaction states with timeout handling and duplicate protection.
- Built and owned **PRM, the Payment Risk Management Engine**, evaluating **20+ transaction attributes** with Redis-backed state, idempotent decisions and graceful degradation. Integrated **generative AI into reviewable fraud investigations**, contributing to roughly **35% fewer fraud-related production incidents**.
- Decoupled processing across **5+ services using Kafka and Azure Service Bus**, with partition routing, consumer isolation, retries and recovery. Bounded asynchronous pipelines, cancellation and backpressure kept intake responsive at **5,000+ concurrent connections per node**.
- Reduced **P99 latency by roughly 30%** through Redis caching, deduplication, distributed locks and TTL policies; tuned database access and connection pools to remove recurring bottlenecks.
- Built treasury-liquidity and ledger workflows with **atomic debits and credits** and idempotent recovery; connected **NIBSS, Monnify and Open Banking**. Provider validation, retries and timeouts across **AML/KYC integrations** cut manual compliance review by roughly **40%**; settlement-reconciliation reports surfaced missing records and exceptions for Operations.
- Delivered payment validation, **LDAP authentication and SMS notifications** for **10 institutions and 30+ production endpoints**. Protected interbank communication with TLS/mTLS, X.509 and signing; resolved **90% of identified high-severity Veracode findings** in reviewed services and added regression protection.
- Owned **AWS ECS/EC2 deployments** and investigated stalled payments across logs, traces, messaging and persistence. Added deterministic regression tests, production-traffic replay and **Prometheus/Grafana** alerts so recurring failures became reproducible cases with corrective fixes.
- Contributed to a **LiteLLM AI gateway for 50+ engineers**, with tenant limits and model routing, and an onboarding assistant that reduced onboarding time by roughly **25%**. Built a **Claude Code/Codex workflow** with reusable skills, project context, TDD and mutation testing; verified generated changes through review, security checks and runtime evidence.

Tech stack: Java | C#/.NET | Python | Kafka | Azure Service Bus | Redis | gRPC | PostgreSQL | AWS ECS/EC2 | Azure | Kubernetes | LiteLLM | Azure OpenAI | OpenTelemetry | LGTM (Loki, Grafana, Tempo, Mimir) | SonarQube | Veracode

Senior Software Engineer *Jun 2024 - Apr 2025*

Huawei Technologies Remote

Partner & Enterprise B2B Platform

- Delivered a **partner rewards administration platform** from requirements through release, with transactional APIs, asynchronous provisioning, tenant isolation and auditable access controls; supported rollout across **5 African markets: Nigeria, Ghana, Kenya, Cameroon and South Africa**.
- Moved **email, SMS and push delivery** off synchronous request paths through a shared **RabbitMQ notification service across 5+ services**. Idempotent consumers, failover and dead-letter retries isolated delivery failures from business operations and sustained **99% uptime**.

- Led migration of **250,000+ live Redis keys** to **Amazon ElastiCache Serverless for Valkey**. Used redis-shake for live replication and synchronised outstanding changes before cutover, preserving sessions with **zero user logouts and zero downtime**.
- Standardised **Azure DevOps, Helm and Terraform** delivery with automated integration checks and release quality gates, achieving **25% shorter deployments**, a **95% bug-free release rate** and **20% faster delivery**.

Tech stack: Java | C#/.NET | Kafka | RabbitMQ | Redis | AWS | Azure DevOps | Kubernetes | Helm | Terraform | Testcontainers

Software Engineer (Contract) *Jan 2024 - Jun 2024*
Vision Forge AI Automations Remote, Nigeria
AI Workflow & Integration Platforms

- Built the **Intertrade approval workflow engine**, automating multi-step sign-off and approval routing. Connected **5 distributed services through RabbitMQ** with validation and idempotency, reducing **QA data errors by 50%**.
- Moved long-running integrations into **background workers** with bounded producer/consumer pipelines, cancellation and resilient outbound calls, reducing **average response time by 15%**.
- Designed **GraphQL/Apollo contracts with DataLoader batching** across heterogeneous service data sources, giving operational workflows a consistent integration surface.
- Secured multi-tenant workflows with **OAuth 2.0/JWT and RBAC** and worked with QA to expand automated regression protection to **85%+ coverage** using containerised integration tests and provider mocks.

Tech stack: Java | C#/.NET | Node.js | RabbitMQ | GraphQL | Apollo | PostgreSQL | OAuth 2.0 | Testcontainers | WireMock

Software Engineer *Jan 2023 - Dec 2023*
Schlumberger (SLB) Oil & Gas Servicing Limited Lagos, Nigeria
Enterprise Applications & Integrations

- Built a **wellsite-data ingestion service using Kafka and PostgreSQL**, validating equipment events and handling duplicate, delayed and out-of-order messages. Retries, replay and backpressure kept downstream processing recoverable during connectivity interruptions.
- Delivered **REST APIs and PostgreSQL/SQL Server models** for operational workflows, with request validation, rate limiting and resilient outbound integrations; protected sensitive operations with **OAuth/OIDC, JWT, RBAC and Twilio two-factor authentication**.
- Architected **Azure/GCP delivery pipelines** and right-sized workloads, reducing **cloud costs by 18%**.
- Reviewed architecture and production changes with users, QA and infrastructure teams; investigated API, persistence and provider failures, documented corrective actions and supported releases through stabilisation.

Tech stack: Java | C#/.NET | Kafka | PostgreSQL | SQL Server | Azure | GCP | OAuth 2.0 | Twilio

Software Engineer *Jan 2022 - Dec 2022*
The Intrepid Technologies (Chevron) Lekki, Nigeria
Enterprise Applications & Automation

- Removed API and database bottlenecks across **3 enterprise services** through query-plan analysis, indexing and connection pooling, reducing **average response time by 200ms** and increasing **database throughput by 35%**.
- Reworked **L1/L2 caching, Redis pub/sub and gRPC communication**, consolidating large entities and tuning data-access paths to reduce infrastructure and database costs while preserving production performance.
- Enhanced batch jobs with execution metrics and automated purging of **500,000+ expired records daily**; automated operational reporting to reduce **reporting-cycle time by 40%**.
- Raised automated coverage from **20% to 85%** through TDD/BDD, containerised integration tests and regression cases. Used **Prometheus, Grafana, Splunk and CloudWatch** for incident investigation, SLA monitoring and corrective actions.

Tech stack: Java | C#/.NET | Oracle | PostgreSQL | Redis | gRPC | AWS CloudWatch | Prometheus | Grafana | Splunk | Testcontainers

Software Engineer (Contract) *Jan 2020 - Dec 2021*
Upwork Remote
Backend Services & Client Integrations

- Designed **versioned REST APIs and third-party integrations** with explicit timeouts, idempotent handlers, connection pooling and structured logs; documented contracts and validated failure paths with client Product, QA and Operations teams.
- Kept external work off customer-facing request paths using **asynchronous processing, AWS Lambda, Amazon SNS and SQS**. Published account-lifecycle events for retryable notifications and background work without blocking core operations.
- Owned **AWS deployment pipelines and Linux application environments**, including release verification, monitoring, incident investigation and operational handover, so clients could maintain and troubleshoot the services after delivery.

Tech stack: Java | C#/.NET | Node.js | Kafka | AWS | Lambda | SNS | SQS | Linux | REST

Selected Projects

IRIS - Core Banking Payment Gateway *(Internal)* - TCP, gRPC/Protobuf, Kafka

- Built **ISO 20022/SWIFT validation and recovery paths** covering message framing, signatures, canonicalisation, idempotency and deterministic regression cases for duplicate, delayed and failed messages.

PRM - Payment Risk Management Engine *(Internal)* - Redis, CQRS, generative AI

- Separated **20+ transaction-attribute checks** from payment services; integrated **GenAI fraud-investigation workflows** while keeping transaction decisions deterministic, failure-tolerant and auditable.

StableX - Wallet & Hybrid Exchange Prototype **[Wallet Demo]** - Go, Solidity, Hardhat

- Built a **Go order book and local-chain Solidity AMM** with exact accounting, reserved balances, idempotent order IDs, partial fills and recovery; verified failure paths through **Go concurrency/API tests, 10 Solidity tests and browser checks**. Paper balances and local-chain tokens remain isolated from custodial funds.

Education

Babcock University Nigeria

Bachelor of Science (Honours) in Software Engineering - GPA: 4.10/5.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Software Architecture

Certifications

- Oracle Cloud Infrastructure DevOps Certificate, Oracle (Oct 2025)
- Oracle Cloud Infrastructure Developer Certificate, Oracle (Oct 2025)
- Oracle 12c Administrator / Java SE 8 (OCA/OCP), New Horizons (Jul 2024)
- Domain 4 Review: AWS Certified Solutions Architect - Professional (SAP-C02), AWS
- AWS Fundamentals of Architecting on AWS, AWS
- Microsoft Applied Skills: Deploy Cloud-Native Apps Using Azure Container Apps, Microsoft
- Microsoft AI Agentic Cloud Developer Certificate, Microsoft
- Microsoft AZ-204: Developing Solutions for Microsoft Azure (In Progress)
- UiPath RPA Certification, UiPath (Jun 2023)
- Udemy Java Masterclass, Udemy (Jan 2023)

Awards & Affiliations

- **3rd Place - Wema Bank Hackathon (Dec 2025):** NeuraFeed, an AI-driven platform delivering personalised financial insights.
- Student Lead, Enactus (Aug 2023) - Digital Onboarders, MindTheGap (Jan 2024) - IEEE Graduate Member